



Prognostic Effectiveness of PD-L1 Tumoral Expression in Oral Cavity Squamous Cell Carcinoma

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Abstract

PD-L1 is a glycoprotein present on antigen-presenting cells and malignant cells and activates the PD-L1 receptor on cells surfaces, leading to T-cell anergy and death. The objective of this study was to assess PD-L1 tumoral expression in 50 patients with oral squamous cell carcinoma, define its prognostic value, and investigate its association with 2-year overall survival, locoregional recurrence, distant metastasis, and a number of other clinicopathological parameters. In a 24-month prospective observational study, involving 50 oral cavity squamous cell carcinoma patients, PD-L1 tumoral expression was evaluated using semiquantitative immunohistochemistry analysis by an experienced pathologist. PD-L1 expression of $\geq 6\%$ of tumor cells was associated with poor survival outcome and determined to be a pertinent clinical cutoff value. PD-L1 expression of $\geq 6\%$ was significantly associated with a number of clinicopathological parameters in our study of 50 patients with OSCC. Pathological nodal staging ($P=0.00$), lympho-vascular invasion ($P=0.03$), extra-nodal extension ($P=0.04$), overall staging ($P=0.01$), locoregional recurrence ($P=0.00$), and distant metastasis ($P=0.00$) all showed statistically significant associations. Our study concluded a significant correlation with decreased 2-year overall survival and 2-year disease free survival by using the Kaplan- Meier survival plot for overall survival ($p=0.02$) and ($p=0.0002$), respectively. The presence of $\geq 6\%$ PD-L1 (CD274) tumoral expression was found to be significantly associated with 2-year overall survival (OS), locoregional recurrence (LRC), distant metastasis (DM), and various clinicopathological parameters. Tumoral PD-L1 was found as a discrete prognostic biomarker which showed significant association with tumor aggressiveness.

Keywords PD-L1 tumoral expression · Oral squamous cell carcinoma · Head and neck cancers · Prognostic biomarker · Overall survival · Locoregional recurrence · Distant metastasis · Prospective study

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Introduction

Oral cavity cancer (OCC) accounts for 2% to 4% of all cancer cases worldwide, ranking as the sixth most prevalent malignancy. India bears the largest burden of OCC cases, accounting for approximately one-third of worldwide occurrences. The majority of OCC cases are oral squamous cell carcinomas (OSCC), which make up 90% of all OCC cases [1]. In India, an estimated 52,000 deaths and 77,000 new cases are recorded annually. The rising incidence of OCC is a major concern in India community healthcare system. Risk factors for OSCC include tobacco use (smoker and smokeless), betel nut, alcohol consumption, and HPV infections.

The standard treatment for OSCC is surgery, with adjuvant radiotherapy or concurrent chemoradiotherapy (CTRT) frequently employed for advanced stage disease and in the presence of adverse features.

OSCC is an immunosuppressive cancer. The PD-L1 receptor on thymus cells, bone marrow cells, and macrophages [2]. PD-L1 is a glycoprotein found on cells surfaces that is present on APC (antigen-presenting cells) along with malignant cells that activate their PD-L1 receptor to cause T-cell anergy and death. Within the tumor microenvironment, PD-L1 pathway controls the establishment and maintaining of immunological tolerance. T cell activation, proliferation, and cytotoxic secretion in cancer are caused by the action PD-L1; this results in the decline of anti-tumor immune reactions [3]. Yu and colleagues [4] in a metanalysis confirmed that patients of HNSCC had considerably more levels of the mRNA descriptor and gene ploidy of Cluster of differentiation 274, which encodes programmed death ligand-1 receptor respectively.

Clinical efficacy has been demonstrated for PD-1 checkpoint inhibitors, such as pembrolizumab and nivolumab, in the treatment of advanced and recurrent HNSCC [5, 6]. PD-L1 expression, assessed through immunohistochemistry, is believed to predict clinical response [7]. A range of malignancies, including skin, kidney, upper gastrointestinal, breast, and brain tumors, also have increased PD-L1 expression [8] and has been linked to a poor prognosis in various cancers, including melanoma and carcinomas of the salivary glands, nasopharynx, upper gastrointestinal, kidney, and breast [9–12].

Inhibition of the PD-L1 pathway may hold promise for reducing tumor development and improving survival outcomes. However, the clinical significance and prognostic relevance of PD-L1 in squamous cell carcinoma of the oral cavity remains unclear, as there have been limited studies in this area.

Material and Methods

A prospective observational study was done at the Department of Head and Neck Surgery and Pathology on 50 patients with OSCC, AIIMS, Raipur, Chhattisgarh, during a period of February 2021 to March 2023, after getting acquiescence from the IEC (Institutional Ethics Committee) of AIIMS Raipur, approval number 1467/2021 for the project 719/2021.

Study Population

The sample size for this study was calculated by standard formula [13], patients with histo-pathologically confirmed cases of OSCC who had surgery at the Department of Head & Neck Surgery, AIIMS Raipur, Chhattisgarh, and all the patients fulfilling the inclusion criteria were included in the study. Standard recommendations were followed for patients who underwent surgery with or without reconstruction

and adjuvant therapy (radiotherapy/chemoradiotherapy) if required as per the protocol, without changing the specimen's orientation or margin status; the specimen was delivered to the Department of Pathology for histopathological analysis after surgery.

The most relevant tumor region (tumor core) was identified after microscopic analysis. The necessary immunohistochemistry was carried out in the chosen blocks.

An expert pathologist performed the specimen histopathological evaluation. PD-L1 antibody clone 405-9A11 by DBS was used. Immunohistochemistry (IHC) for Programmed Death- Ligand 1 receptor (PD-L1-CD274) was performed on the paraffin-embedded blocks as per the standard protocol for immunohistochemical staining in the lab. An experienced pathologist manually examined immuno-staining.

The cytoplasmic PD-L1 expression was analyzed for both the tumor core and invasive margin tumor cells.

PD-L1 expression: scoring criteria:

- 1) Score 0: $\leq 1\%$ positivity of tumor cells
- 2) Score 1: 2–5%
- 3) Score 2: 6–10%
- 4) Score 3: $> 10\%$ [14]

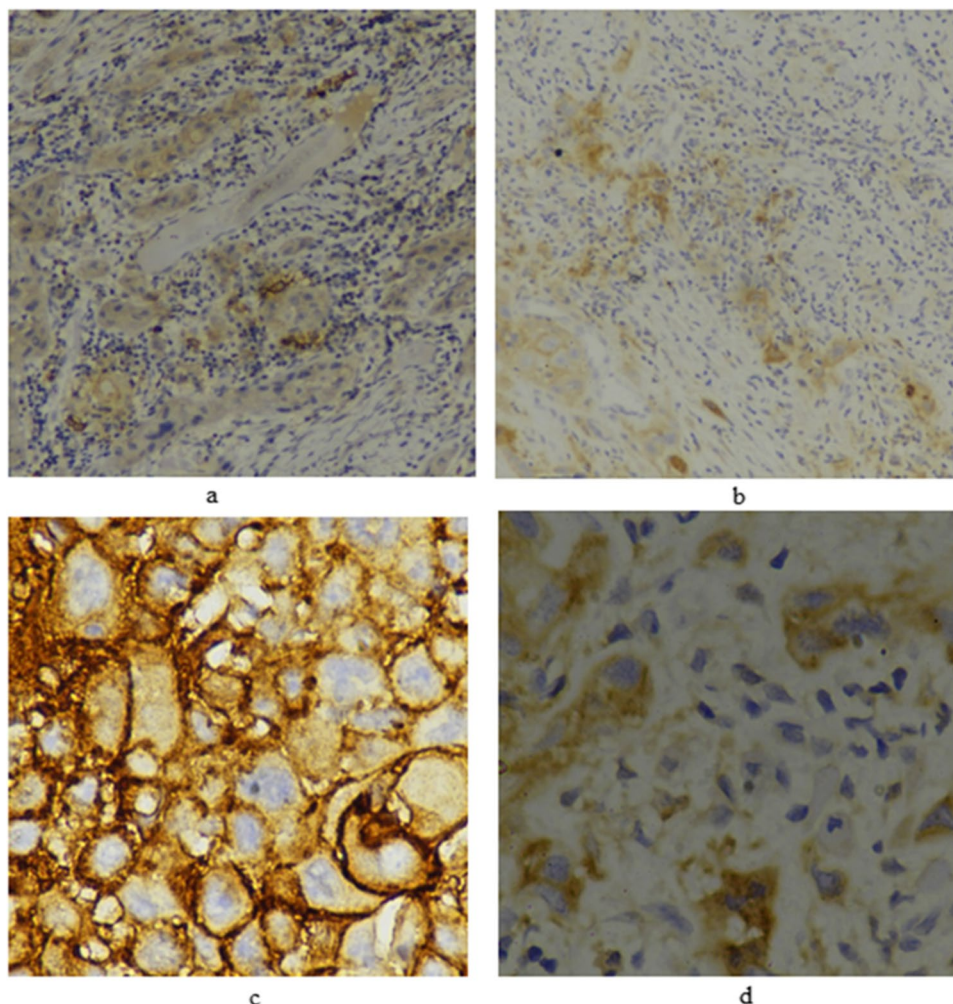
The tumoral expression of PD-L1 of $\geq 6\%$ was associated with poor outcome in our analysis, so it was established as cut off criteria for further study (Fig. 1).

Statistical Anatomization

Data was transcribed into a Microsoft Excel (ME), and statistical anatomization was performed availing SPSS adopting 27.0, SPSS Inc. and GraphPad Prism adopting 5th version. Counts and percentages of the data had been compiled for categorical vacillating, mean, and standard deviation for numerical vacillating. Independent samples or unpaired samples were used in two-sample t-tests to see if the means were different. Blocking techniques such as paired t-tests had more power than unpaired tests. Any statistical hypothesis test where the test statistics sampling dispensation is a chi-squared dispensation when the null hypothesis is true is known as a chi-squared test. Unmatched/unpaired expanse was examined using the relevant Chi-square (test) or with a Fischer's exact test.

Correlation was determined using the Pearson correlation analysis. The linear dependence between two variables X and Y was scaled by the Pearson outcome-moment correlation quantum. Below is precise guise that can be owned to perform diverse t-tests. The formula for a test statistic that closely resembles a t-distribution test under the null hypothesis in each instance is provided. In each scenario, the appropriate degrees of freedom are also provided. Each

Fig. 1 Photomicrograph of PD-L1 expression (20× and 40× magnification)



of these statistics can be used to carry out either a one tailed test or two tailed test. The Kaplan–Meier curve was used to do a study of overall survival (OS), and the difference in survival was examined using the Log-Rank test.

Utilizing a merit of values from Student's *t*-distribution, one can find a *p* value using a table of values after determining a *t* value. The *p* value is used to adjudicate the level of statistical significance with a threshold of <0.05 being used.

Results

An aggregate of 50 patients were incorporated in this study. In our analysis, mean age of presentation was 45.8800 (± 10.5997). In our study (Table 1), 10 (20.0%) patients were female, and 40 (80.0%) patients were male. Patient mostly presented with Eastern Cooperative Oncology Group (ECOG) performance status of 0 (94%). Most patient had addiction of tobacco chewing (64%). Carcinoma buccal mucosa was the most common subsite (46%). Among the patients, 10 (20%) had T1 disease, 27 (54%) had T2 disease,

6 (12%) had T3 disease and 7 (14%) had T4 disease, and 56% had advanced stage disease (stage III and stage IV). In our study, G2 (66%) was the most common histological grading, maximum patients had clear margin (76%), 12% patients had perineural invasion, 22% had lympho-vascular invasion, 28% patients had locoregional recurrence, and 12% patients had distant metastasis. Adjuvant treatment was administered to 68% of patients. Over a period of 24 months follow-up (median 17 months), 5 deaths occurred.

Association of PD-L1 Expression with Clinicopathological Parameters

In our analysis, we found a statistical significance between PD-L1 and several clinicopathological parameters, using a cutoff of 6% PD-L1 tumoral expression as an indicator of poor outcome (Table 2). Among the patients in our analysis, 24 (48%) had $<6\%$ PD- L1 tumoral expression, and 26 (52%) had $\geq 6\%$ PD- L1 tumoral expression. A statistically significant association with pathological nodal staging was found in our study ($P=0.00$). Similar statistically significant

Table 1 Apportion of clinicopathological parameters

		Frequency	Percentage
Gender	Male	40	80
	Female	10	20
ECOG performance status	0	47	94
	1	3	6
	2	0	0
Charlson comorbidity index	2	16	32
	3	12	24
	4	6	12
	5	0	0
	6	13	26
	7	1	2
	8	1	2
	9	1	2
Addiction	Alcohol and smoker	2	4
	Areca nut	0	0
	No addiction	1	2
	Tobacco	33	66
	Tobacco and alcohol	8	16
	Tobacco and smoker	1	2
	Tobacco, smoker and alcohol	5	10
Premalignant lesion	No	20	40
	Yes	30	60
Subsites	Ca buccal mucosa	23	46
	Ca lower alveolus	3	6
	Ca lower GBS	1	2
	Ca lower lip	1	2
	Ca retromolar trigone	2	4
	Ca tongue	17	34
	Ca upper alveolus	2	4
	Ca upper lip	1	2
Laterality	Left	26	52
	Right	24	48
Pathological primary staging	pT1	9	18
	pT2	26	52
	pT3	9	18
	pT4a	6	12
Pathological nodal staging	pN0	31	62
	pN+	17	34
	pNx	2	4
Overall staging	Early stage (I and II)	22	44
	Advanced stage (III and IV)	28	56
Histological grading	G1 (WDSCC)	15	30
	G2 (MDSCC)	33	66
	G3 (PDSCC)	2	4
Depth of invasion (cm)	<0.5	23	46
	0.6–1.0	16	32
	>1.0	11	22
Margin status	Clear	38	76
	Close	12	24
	Positive	0	0

Table 1 (continued)

		Frequency	Percentage
Pathological Bone involvement	No	46	92
	Yes	4	8
Pathological skin involvement	No	45	90
	Yes	5	10
Perineural invasion	No	44	88
	Yes	6	12
Lympho-vascular invasion	No	39	78
	Yes	11	22
Extra-nodal extension	No	46	92
	Yes	4	8
Locoregional recurrence	No	36	72
	Yes	14	28
Distant metastasis	No	42	84
	Yes	8	16
Outcome	Alive without disease	36	72
	Alive with disease	9	18
	Dead	5	10
PD-L1 Score	0	7	14
	1	17	34
	2	14	28
	3	12	24
PD-L1 expression	<6% +	24	48
	≥6% +	26	52
Adjuvant treatment	Yes	34	68
	No	16	32

associations were found for lympho-vascular invasion ($P=0.03$), extra-nodal extension ($P=0.04$), overall staging (0.01), locoregional recurrence (0.00), and distant metastasis (0.00). There was no other clinicopathological factor that was statistically significantly linked to PD-L1 expression.

Impact of PD-L1 Tumoral Expression on Survival of Patient

In our analysis, over a follow-up period of 24 months, we observed a statistically significant association with disease free survival (DFS) ($p=0.0002$) and overall survival (OS) ($p=0.02$). Among the patients who had PD-L1 expression levels $\geq 6\%$, 13 patients (50%) had recurrence at 24 months, and 1 (4.16%) patient had recurrence at 24 months among the patients who had PD-L1 expression level $< 6\%$. DFS along with PD-L1 expression using Kaplan- Meier survival plot is shown in Fig. 2. DFS at 2 years was 95.84% and 50% for PD-L1 expression $< 6\%$ and $\geq 6\%$, respectively, which was found to be statistically significant ($p=0.0002$). Only one patient had PD-L1 expression that was $< 6\%$, and eight patients who were still alive with the disease had PD-L1 expression that was $\geq 6\%$. All five patients who died had $\geq 6\%$ PD-L1 expression. Our research revealed

that PD-L1 expression was $\geq 6\%$ in all deceased patients (Table 3) (Fig. 3).

Discussion

This purpose of the study was to determine the expression of tumoral PD-L1 and its association to (OS) overall survival, locoregional recurrence (LRC), distant metastasis (DM), and other clinicopathological parameters in 50 patients. It also sought to determine the prognostic significance of these findings. Ishida [15] originally identified PD-L1 as an immunoinhibitory receptor in 1992. Tumor cells as well as several other types of immune cells, such as activated Bone marrow (B), thymus cells (T) cells, and various other significant cells like macrophages, and dendritic cells also express PD-L1.

A pivotal immune system regulator, PD-L1, is very critical for the immunological evading mechanisms of numerous variable malignant cells [16]. In certain solid tumor, overexpression of PD-L1 creates a critical immunosuppressive tumor microenvironment and hinders T lymphocytes from mediating cytolysis. Various noted studies have demonstrated the correlation between clinicopathological factors and outcome/prognosis.

Table 2 Association of PD-L1 expression with clinicopathological parameters

		PD-L1 expression (%)		<i>P</i> value
		<6% +	≥6% +	
Gender	Male	19	21	0.88
	Female	5	5	
Premalignant lesion	No	11	9	0.30
	Yes	13	17	
Pathological primary staging	pT1	4	5	0.96
	pT2	12	14	
	pT3	5	4	
	pT4a	3	3	
Subsites	Ca buccal mucosa	13	10	0.64
	Ca lower alveolus	1	2	
	Ca lower GBS	1	0	
	Ca lower lip	1	0	
	Ca retromolar trigone	1	1	
	Ca tongue	6	11	
	Ca upper alveolus	1	1	
	Ca upper lip	0	1	
Pathological nodal staging	pN0	22	9	0.00
	pN +	1	16	
	pNx	1	1	
Overall staging	Early stage (I and II)	15	7	0.01
	Advanced stage (III and IV)	9	19	
Histological grading	G1 (WDSKC)	6	9	0.75
	G2 (MDSKC)	17	16	
	G3 (PDSKC)	1	1	
Depth of invasion (cm)	< 0.5	12	11	0.85
	0.6–1.0	7	9	
	> 1.0	5	6	
Lympho-vascular invasion	No	23	16	0.03
	Yes	1	10	
Extranodal extension	No	24	22	0.04
	Yes	0	4	
Perineural invasion	No	21	23	0.91
	Yes	3	3	
Locoregional recurrence	No	23	13	0.00
	Yes	1	13	
Distant metastasis	No	24	18	0.00
	Yes	0	8	
Status of patients at the end of follow-up	Alive with-out disease	23	13	0.02
	Alive with disease	1	8	
	Dead	0	5	

Significant *p* value

Yagyuu et al. [17] concluded that expression of PD-L1 in subepithelial or epithelial premalignant lesions are linked to the malignant transformation.

In our study of 50 patients with OSCC, PD-L1 expression was significantly associated with several clinicopathological parameters. Statistically significant associations were found

for pathological nodal staging ($P=0.00$), lympho-vascular invasion ($P=0.03$), extra-nodal extension ($P=0.04$), overall staging (OAS) (0.01), locoregional recurrence (LRC) (0.00), distant metastasis (0.00), and with overall survival (0.02).

Our study showed a significant correlation with factors associated with decrease or poor outcome like locoregional

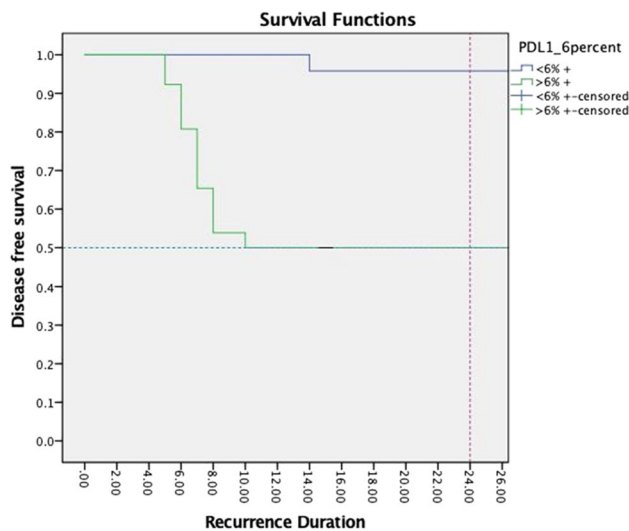


Fig. 2 Depiction of disease-free survival along with PD-L1 expression using Kaplan–Meier survival plot

Table 3 Association of overall survival by using Log-Rank test

Dead or alive at the time of follow-up		Alive	Dead	<i>P</i> value
PD-L1 expression	<6% +	24	0	0.02
	≥6% +	21	5	

Significant *p* value

recurrence ($P=0.00$) and distant metastasis ($P=0.00$). Similar result was found by Dong-Ho Geum et al. [18]; the study revealed statistically significant correlation between variables like metastasis and locoregional recurrence. Lin et al. [19] also concluded that PD-L1 expression in oral carcinoma (OSCC) was associated to distant metastasis.

Our study showed a significant association of PD-L1 with nodal metastasis ($P=0.00$), similar association was found

by Dong-Ho Geum et al. [18] which also showed significant correlation with nodal metastasis. Another study by Yong-Xin Cui et al. [20] also concluded that increased PD-L1 tumoral expression was significantly associated with differentiation of tumor (G1/G2/G3) and lymph node metastasis.

In our study, PD-L1 tumoral expression was significantly associated with (LVI) lympho-vascular invasion (p value 0.03). Similar study by Vijay et al. [21] also revealed a significant association to lympho-vascular invasion. If nodal metastasis also considered along with LVI, both contribute to the disease's aggressiveness.

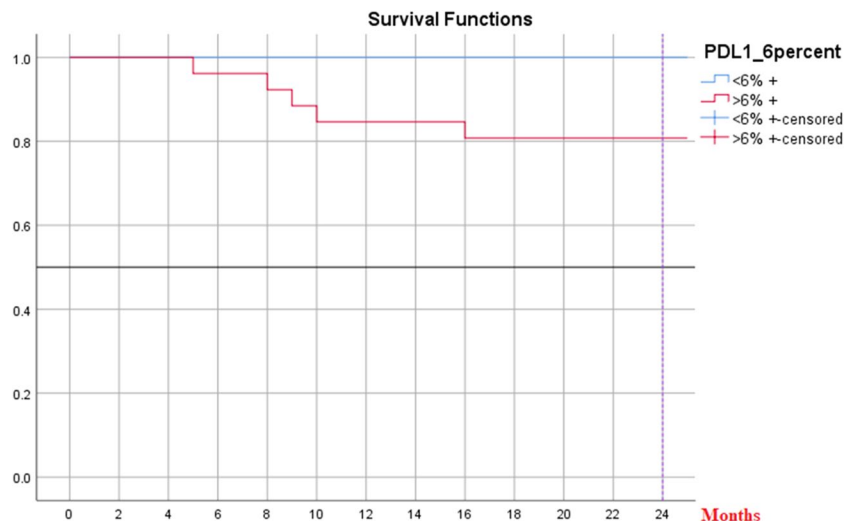
Geum et al. [18] exhibited statistical significance according to tumor size and TNM stage, which was consistent with our findings of statistical significance with overall staging ($P=0.01$). Cui et al. [20] concluded that grading of tumor was statistically correlated with high PD-L1 expression, which is inconsistent with our results.

Among other clinical and pathological variables studied, our study did not show any significant association with gender, tumor subsite, histological grading of tumor, perineural invasion, and depth of invasion. One of the studies done by Takamaru et al. [22] also demonstrated that no association with sex was observed, but another study done by Lin et al. [19] observed that higher levels of PD-L1 expression in OSCC were significantly correlated with female sex.

In our analysis, we also found that PD-L1 tumoral expression significantly reduced overall survival in patients, using a criteria of $\geq 6\%$ PD-L1 expression. All five patients who died had PD-L1 expression of $\geq 6\%$ expression in tumor cell.

According to the inference clinched from our study, PD-L1 positivity in OSCC was statistically significant and associated with nodal positivity, lympho-vascular invasion, extra-nodal extension, locoregional recurrence, distant metastasis, and overall survival. Each of the aforementioned factors are linked to disease aggression and negative outcomes.

Fig. 3 Depiction of overall survival along with PD-L1 expression using Kaplan–Meier survival plot



Even though the exact mechanism underlying PD-L1 overexpression is still not completely grasped, however, a recent meta-analysis by Gandani and colleagues [23] found a correlation between IHC assessment of tumor PD-L1 expression and clinical response to PD-L1 immunotherapy in different cancers. Topalian et al. [24] concluded that antiPD-L1 targeted therapy was effective in treating tumors that were PD-L1 positive.

Limitation

The limited sample size and the shorter follow-up were found to be one of the limitations of our study.

Future Direction

Further research focused on PD-L1 expression could provide additional valuable information and insight into the impact of this marker on the tumor microenvironment. These findings may contribute to the treatment of OSCC and the probable addition of immunotherapy along with aggressive follow-up. However, the limited number of prospective studies and the non-uniform population have hindered the literature. Our study highlights the autonomous role of PD-L1 in the prognosis of oral squamous cell carcinoma. To support our findings, larger studies with a larger sample size and longer follow-up periods are needed to establish tumoral PD-L1 as a discrete prognostic biomarker.

Conclusion

For patients with OSCC, expression of PD-L1 (CD274) is the statistically significant prognostic value that has been demonstrated. The presence of $\geq 6\%$ PD-L1 expression was found to be significantly associated with various clinicopathological parameters, 2-year overall survival and 2-year disease free survival. PD-L1 was found to be an independent prognostic marker, showing a significant association with tumor aggressiveness. PD-L1 expression in the tumor cells can be considered a predictor of an unfavorable outcome, as cancer cells escape immune surveillance.

Declarations

Conflict of Interest The authors declare no competing interests.

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